Introduction to Machine Learning

Machine learning is a branch of artificial intelligence that enables computers to learn patterns from data and make predictions or decisions without being explicitly programmed for every case.

Supervised Learning

Supervised learning uses labeled training data. Each training example contains an input and a known target output. Common supervised learning tasks include classification and regression. Classification predicts a category, while regression predicts a numerical value.

Unsupervised Learning

Unsupervised learning works with data that does not have target labels. Clustering is a common example. A clustering algorithm groups similar data points together based on their characteristics.

Model Evaluation

Evaluation measures how well a machine learning model performs on unseen data. For classification, accuracy, precision, recall, and F1-score are commonly used. Accuracy is the proportion of correct predictions among all predictions.

Overfitting

Overfitting happens when a model learns the training data too closely and performs poorly on new data. Regularization, simpler models, and more training data can help reduce overfitting.